

Glimpse: Quantile Prediction Markets on Bitcoin

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Abstract

Existing prediction markets rely on the continuous double auction, a zero-sum mechanism that requires noise traders for liquidity, permits profitable bluffing, and fragments liquidity across outcome spaces. We present the Quantile Liquidity-Sensitive Logarithmic Market Scoring Rule (QLS-LMSR), an automated market maker that partitions the Bitcoin price axis into n quantile bins and prices all outcomes through a single convex cost function. The mechanism is positive-sum: the operator subsidizes a bounded, pre-committed worst-case loss, and in return obtains a continuously updating probability distribution over the full outcome space. Pricing is path-independent and liquidity scales endogenously with trading volume. Under the conditionally independent signal structure natural to financial forecasting, truthful betting is the unique Weak Perfect Bayesian Equilibrium, so no trader can profit by bluffing or withholding information. As a running example, we instantiate the mechanism on Bitcoin closing price with 500 bins, denominated and settled in satoshis.

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1 Introduction

Prediction markets consistently outperform experts, polls, and consensus panels at forecasting uncertain events [10, 9, 4]. Yet they have been confined almost entirely to categorical questions—politics, sports, or entertainment—where the outcome space is narrow and the forecasting value is modest. The domain where prediction markets would be most useful is quantitative financial forecasting: not “Will Bitcoin hit \$100,000?” but “Where will Bitcoin close at some point in time in the future?” That question has never been answered by a prediction market, because no existing mechanism can support it.

Nearly every prediction market in operation runs on the continuous double auction (CDA). The CDA is a zero-sum game, and rational agents in a zero-sum market will not trade at all, because any offer reveals information that discourages the counterparty [14]. Trading occurs only when noise traders—participants who trade for non-informational reasons—supply the other side, which makes information aggregation contingent on the presence of uninformed capital [13]. Even then, informed participants will leak their signals gradually to maximize profit, and under certain information structures they can profitably bluff [7, 8, 3].

To produce a coherent forecast over a quantitative outcome space, we propose the Quantile Liquidity-Sensitive Logarithmic Market Scoring Rule (QLS-LMSR): a single automated market maker that partitions the price axis into n bins and prices all of them through one convex cost function. The operator commits a fixed subsidy, and in return every trade updates a complete probability distribution over the outcome space. Liquidity is guaranteed from the first trade, with no dependence on noise traders or matched counterparties. Prices are path-independent, so the operator’s worst-case loss is bounded and determined before the market opens. Under the signal structure natural to financial forecasting—where each trader’s private information is drawn independently conditional on the true outcome—truthful reporting is the unique Weak Perfect Bayesian Equilibrium, so no trader can profit by misrepresenting beliefs [12, 8].

We apply the mechanism to Bitcoin closing price, using 500 bins as a running example. Bitcoin is a natural first application because its market structure satisfies the mechanism’s key assumption without approximation. It trades globally and continuously with no privileged disclosure cycle—no earnings reports, no management guidance, no insider windows—so traders estimate the closing price from independent data using independent methods. This is precisely the conditionally independent signal structure under which truthful trading is the unique equilibrium. The diversity of participants—spanning jurisdictions, sophistication levels, and ideological orientations—is a feature: each independent signal refines the forecast. Settlement occurs natively in satoshis, requiring no jurisdictional payment intermediary. The result is a complete, continuously updating probability distribution over where Bitcoin will close, produced and refined by every trade.

2 Background Information

A *prediction market* is a mechanism for aggregating dispersed information about an uncertain event whose outcome will be objectively verifiable at a future time. The market offers state-contingent claims (event contracts) and uses trading activity to produce an interpretable public forecast, represented by prices that reflect market consensus probabilities. This mechanism rests on a mathematical foundation of scoring rules, which were originally developed to evaluate and reward accurate probabilistic forecasts in domains like meteorology. Proper scoring rules create incentives for forecasters to report their true beliefs—a principle that extends to market scoring rules, which adapt the concept to sequential trading among many participants (See Section 7 for more detail). In this setting, trade is a means to an informational end: the operator rationally subsidizes participation and liquidity in order to obtain better forecasts. To understand how these incentives are formally constructed, we begin with the theory of proper scoring rules and their extension to market scoring rules.

2.1 From Probabilistic Forecasting to Market Scoring Rules

The conceptual origin of prediction markets lies in *probabilistic forecasting*. Long before electronic markets, forecasters in domains such as meteorology were asked to report probabilities of events (e.g., precipitation), and these probability forecasts were evaluated using *scoring rules*. A scoring rule is a contract that assigns a numerical score (or monetary payoff) to a reported distribution $r = (r_1, \dots, r_m)$ once the realized outcome is known. Early work by Brier introduced a quadratic scoring rule for verifying probability forecasts in weather forecasting, and subsequent work formalized the logarithmic scoring rule, with both families becoming standard tools for evaluating probabilistic predictions. Scoring rules have since been widely used in weather forecasting and related forecast-verification settings [2, 12].

Formally, let $\Omega = \{1, \dots, m\}$ denote mutually exclusive outcomes of an event. A scoring rule is a collection of functions $\{s_i(\cdot)\}_{i=1}^m$ where $s_i(r)$ is the score paid if outcome i occurs. The rule is (*strictly*) *proper* if a risk-neutral forecaster with true belief p maximizes expected score by reporting $r = p$ [12].

Two canonical examples illustrate this framework. The quadratic rule (Brier score) is given by

$$s_i(r) = a_i - b(1 - r_i)^2, \quad b > 0,$$

where the forecaster receives maximum score when $r_i = 1$ for the realized outcome i and is penalized quadratically for deviations from perfect accuracy. The logarithmic rule takes the form

$$s_i(r) = a_i + b \log r_i, \quad b > 0,$$

which rewards accurate probability estimates through the natural logarithm and becomes arbitrarily punitive as reported probabilities approach zero for the realized outcome. Properness

matters because it creates a disciplined incentive: under standard assumptions, the forecaster is rewarded for stating the probability distribution they actually believe. A concrete illustration of the incentive logic is deferred to Appendix 7, which derives truth-telling under the quadratic (Brier) and logarithmic scores and contrasts it with a simple non-proper rule.

A limitation of a one-shot scoring rule is that it elicits a forecast from *one* forecaster at *one* time. In many applications, information arrives gradually and is distributed across many participants. What is needed is a mechanism that allows many forecasters to contribute over time, while preserving the incentive logic of proper scoring rules. This is where market scoring rules come into play.

2.2 Hanson’s Insight: The Logarithmic Market Scoring Rule (LMSR)

Hanson introduced *market scoring rules* (MSR) precisely to pool opinions from multiple forecasters in a sequential, shared way [12]. An MSR maintains a *current* market distribution $p \in \Delta_m$. At any time, a trader may replace it by a new distribution p' . If outcome i occurs, the trader receives the incremental score improvement

$$\Pi_i = s_i(p') - s_i(p).$$

This is a “pay for improvement” rule: a participant profits if they move the market forecast in a direction that makes the realized outcome more consistent with their information [6].

A key operational feature is that the predictions *telescope*. Each trader effectively takes over responsibility for the current score and replaces it with a new score. As a result, the market maker is responsible only for the difference between the *initial* distribution p_0 and the *final* distribution reached by the last update. This yields a clean bounded-loss guarantee: the market maker’s worst-case loss is finite and can be expressed directly in terms of the scoring rule [6].

While MSRs are conceptually clear, they do not initially resemble familiar “markets” because participants appear to be editing probability vectors rather than trading contracts. However, MSRs are *equivalent* to a more intuitive implementation: a *cost-function market maker* that offers state-contingent contracts [5, 6].

2.3 Equivalence of MSRs and Cost-Function Market Makers

Market scoring rules and cost-function market makers are mathematically equivalent [5, 6]. In the MSR, a trader edits a probability vector directly, whereas in the cost-function formulation, a trader buys and sells state-contingent contracts priced by the gradient of a convex potential. The two representations produce identical payoffs for every trade, provided market probabilities remain strictly positive. We develop the cost-function formulation here because it is the more natural language for implementation.

Let $\Omega = \{1, \dots, n\}$ be mutually exclusive and exhaustive outcomes as before. A cost-function market maker offers n Arrow-Debreu contracts, where contract i pays \$1 if outcome i is realized and \$0 otherwise. Let q_i denote the total outstanding quantity of contract i across all traders, and let $\mathbf{q} = (q_1, \dots, q_n)$. The market maker maintains a differentiable convex cost function $C : \mathbb{R}^n \rightarrow \mathbb{R}$ that records the total money paid in as a function of the quantity vector. A trader who moves the state from $\mathbf{q}$ to $\mathbf{q}'$ pays $C(\mathbf{q}') - C(\mathbf{q})$; when this quantity is negative, the market maker pays the trader. The instantaneous price of contract i is the partial derivative

$$p_i(\mathbf{q}) = \frac{\partial C(\mathbf{q})}{\partial q_i}.$$

We define the price function as the gradient of the cost function, which is a necessary and sufficient requirement for path independence: the cost of moving between any two states depends only on the endpoints, not on the sequence of intermediate trades [6]. Path independence eliminates money pumps and removes any strategic incentive to split or batch orders.

The cost function corresponding to the logarithmic scoring rule is

$$C(\mathbf{q}) = b \log \left(\sum_{j=1}^n \exp(q_j/b) \right),$$

where $b > 0$ is a fixed liquidity parameter. Differentiating yields the price function

$$p_i(\mathbf{q}) = \frac{\exp(q_i/b)}{\sum_{j=1}^n \exp(q_j/b)},$$

which is the softmax. Prices are strictly positive, sum to unity, and increase monotonically in q_i —buying contracts on i raises the price of i and lowers all others. To see the equivalence with the MSR explicitly: a trader who moves the quantity vector from $\mathbf{q}$ to $\mathbf{q}'$ pays $C(\mathbf{q}') - C(\mathbf{q})$ on entry and, if outcome i is realized, nets $(q'_i - q_i) - [C(\mathbf{q}') - C(\mathbf{q})]$. For the logarithmic cost function this equals $b \log(p'_i/p_i)$, which is exactly the payoff prescribed by Hanson’s logarithmic MSR.

The bounded-loss guarantee carries over directly. When the market is initialized at the uniform distribution, the operator’s worst-case loss is $b \log n$ —achieved when the final report concentrates all probability on a single outcome. The parameter b thus governs market depth: larger b means less price sensitivity per trade but a larger subsidy. The convexity of C ensures that successive contracts on the same outcome cost progressively more, which is the mechanism by which the market maker adjusts prices in response to demand.

This cost-function LMSR is the workhorse of modern automated prediction markets [11, 12] and the analytical starting point for the liquidity-sensitive extensions that follow.

2.4 A Liquidity Sensitive LMSR

If we let $b(\mathbf{q}) = \alpha \cdot \sum_j q_j$ then the LMSR cost function becomes liquidity sensitive.

$$C(\mathbf{q}) = b(\mathbf{q}) \log \left(\sum_i \exp(q_i/b(\mathbf{q})) \right)$$

Sensitivity to liquidity is desirable because it squares intuitively with the way we would want markets to function: small investments move prices less in thick (liquid) markets than in thin (illiquid) markets. We can take the partial derivative of the cost function to derive the liquidity sensitive price function as follows:

$$\begin{aligned} p_i(\mathbf{q}) &= \frac{\partial C(\mathbf{q})}{\partial q_i} \\ &= \frac{\partial}{\partial q_i} \cdot b(\mathbf{q}) \cdot \log \left(\sum_i \exp(q_i/b(\mathbf{q})) \right) \\ &= \frac{\partial}{\partial q_i} \cdot \alpha \cdot \sum_j q_j \cdot \log \left(\sum_i \exp(q_i/b(\mathbf{q})) \right) \\ &= \alpha \cdot \log \left(\sum_j \exp(q_j/b(\mathbf{q})) \right) + \frac{\sum_j q_j \exp(q_i/b(\mathbf{q})) - \sum_j q_j \exp(q_j/b(\mathbf{q}))}{\left(\sum_j q_j \right) \cdot \left(\sum_j \exp(q_j/b(\mathbf{q})) \right)} \end{aligned}$$

Therefore, the liquidity-sensitive LMSR price functions is:

$$p_i(\mathbf{q}) = \alpha \cdot \log \left(\sum_j \exp(q_j/b(\mathbf{q})) \right) + \frac{\sum_j q_j \exp(q_i/b(\mathbf{q})) - \sum_j q_j \exp(q_j/b(\mathbf{q}))}{\left(\sum_j q_j \right) \cdot \left(\sum_j \exp(q_j/b(\mathbf{q})) \right)}$$

where $b(\mathbf{q}) = \alpha \cdot \sum_j q_j$. The mechanism is designed to price contracts between 0 and 1 dollars, so we must slightly alter the it to be able to price contracts and trades in a Bitcoin unit of account.

3 A Bitcoin-Denominated Quantile LS-LMSR

The previous sections established the theoretical foundation: proper scoring rules create incentives for truthful reporting, market scoring rules extend this to sequential multi-trader settings, and the liquidity-sensitive LMSR adapts market depth to participation. We now present the full architecture of Glimpse’s automated market maker, which applies these principles to a *quantile outcome space* over Bitcoin closing price.

The key innovation is this: rather than offering a single binary contract (“Will Bitcoin close above \$80,000?”), Glimpse partitions the entire price axis into n mutually exclusive and exhaustive quantile intervals and operates a single LS-LMSR market maker across all n outcomes simultaneously. The result is a traded probability distribution—a complete forecast of where Bitcoin will close, updated in real time by the collective intelligence of all traders.

3.1 The Quantile Outcome Space

Fix a settlement time T and let P_T denote the realized price of Bitcoin in USD at time T . We partition the price axis into n mutually exclusive and exhaustive intervals, called *quantile bins*:

$$\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$$

where each outcome ω_k corresponds to a price range:

$$\begin{aligned} \omega_1 &: P_T \in [\$0, \$1,000) \\ \omega_2 &: P_T \in [\$1,000, \$2,000) \\ \omega_3 &: P_T \in [\$2,000, \$3,000) \\ &\vdots \\ \omega_k &: P_T \in [$(k-1) \cdot w, $k \cdot w) \\ &\vdots \\ \omega_{n-1} &: P_T \in [$(n-2) \cdot w, $(n-1) \cdot w) \\ \omega_n &: P_T \in [$(n-1) \cdot w, \infty) \end{aligned}$$

where w is the bin width in dollars. The parameters n and w are chosen by the operator to cover the plausible price range at the relevant forecasting horizon; a short-horizon market uses narrower bins than a long-horizon one. As a running example throughout this paper we take $n = 500$ and $w = \$200$, covering $[\$0, \$100,000+)$. Exactly one outcome is realized at settlement.

For each outcome ω_k , the market maker offers an *event contract* that pays a fixed payoff of 100 sats if ω_k occurs and 0 sats otherwise. The price of each contract is determined endogenously

by the cost function, and traders express their beliefs by purchasing bundles of contracts across the quantile bins they consider most likely.

Definition 1 (Event Contract). *An event contract on quantile ω_k is a fully collateralized claim that pays the holder 100 sats at settlement if $P_T \in \omega_k$, and 0 sats otherwise.*

The collection of all n contract prices, normalized to the $[0, 100]$ -sat scale, constitutes a market-implied probability mass function over the price axis. This is the *forecast*: a complete probabilistic statement about where the market believes Bitcoin will close, expressed as a histogram that updates with every trade.

3.2 Market State and Contract Quantities

The state of the market at any point in time is fully described by the *quantity vector*

$$\mathbf{q} = (q_1, q_2, \dots, q_n) \in \mathbb{R}_{\geq 0}^n$$

where q_i is the total number of outstanding contracts on outcome ω_i across all traders. As trades occur, the market progresses through a sequence of states:

$$\mathbf{Q} = [\mathbf{q}_0, \mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_m]$$

where $\mathbf{q}_0$ is the initial state set by the market operator and $\mathbf{q}_m$ is the terminal state when the market closes for settlement.

The market is initialized with a uniform *seed quantity* q_0 on every outcome:

$$\mathbf{q}_0 = (q_0, q_0, \dots, q_0) \in \mathbb{R}^n$$

In the default configuration, $q_0 = 500$ contracts per outcome. This seed serves two purposes: it bootstraps the liquidity parameter b to a nonzero value so that the market can accept its first trades, and it defines the market maker's maximum worst-case loss (the initial subsidy).

3.3 The QLS-LMSR Cost and Price Functions

Since we are denominating our trades in a Bitcoin unit of account (where 1 BTC = ₿100,000,000 = 100,000,000 sats), we modify the LS-LMSR cost function by a factor of 100 so that each contract pays 100 sats upon correct realization.

$$C(\mathbf{q}) = 100 \cdot b(\mathbf{q}) \cdot \log \left(\sum_i \exp(q_i/b(\mathbf{q})) \right)$$

We can take the partial derivative of the cost function to derive the price function as follows:

$$\begin{aligned}
p_i(\mathbf{q}) &= \frac{\partial C(\mathbf{q})}{\partial q_i} \\
&= \frac{\partial}{\partial q_i} \cdot 100 \cdot b(\mathbf{q}) \cdot \log \left(\sum_i \exp(q_i/b(\mathbf{q})) \right) \\
&= \frac{\partial}{\partial q_i} \cdot 100 \cdot \alpha \cdot \sum_j q_j \cdot \log \left(\sum_i \exp(q_i/b(\mathbf{q})) \right) \\
&= 100 \cdot \alpha \cdot \log \left(\sum_j \exp(q_j/b(\mathbf{q})) \right) + 100 \cdot \frac{\sum_j q_j \exp(q_i/b(\mathbf{q})) - \sum_j q_j \exp(q_j/b(\mathbf{q}))}{\left(\sum_j q_j \right) \cdot \left(\sum_j \exp(q_j/b(\mathbf{q})) \right)}
\end{aligned}$$

Therefore, the Bitcoin-denominated quantile liquidity-sensitive LMSR price function is:

$$p_i(\mathbf{q}) = 100 \cdot \alpha \cdot \log \left(\sum_j \exp(q_j/b(\mathbf{q})) \right) + 100 \cdot \frac{\sum_j q_j \exp(q_i/b(\mathbf{q})) - \sum_j q_j \exp(q_j/b(\mathbf{q}))}{\left(\sum_j q_j \right) \cdot \left(\sum_j \exp(q_j/b(\mathbf{q})) \right)}$$

where $b(\mathbf{q}) = \alpha \cdot \sum_j q_j$ and $\alpha = v/(n \cdot \ln(n))$, with $v > 0$ a dimensionless tuning constant.

3.4 Worked Example: Market Initialization

We now walk through the complete initialization of the QLS-LMSR under the default Glimpse configuration ($n = 500$, $v = 2$, $q_0 = 500$), computing the liquidity parameter, cost function, price function, and market-implied probabilities from first principles. The purpose is to show that the uniform-state algebra simplifies dramatically and to make the initial market state fully transparent before any trades occur.

3.4.1 The Initial State

The market opens with a uniform seed quantity on every outcome:

$$\mathbf{q}_0 = \underbrace{(500, 500, \dots, 500)}_{n=500} \in \mathbb{Z}^{500}$$

This is a state of maximum ignorance: the market maker has no information about where Bitcoin will close, so it assigns equal weight to every bin from $[\$0, \$1,000)$ through $[\$499,000, \infty)$.

3.4.2 Step 1: The Liquidity Parameter $b(\mathbf{q}_0)$

The liquidity parameter is $b(\mathbf{q}) = \alpha \cdot \sum_j q_j$, where $\alpha = v/(n \ln n)$. At the uniform state, $\sum_j q_j = n \cdot q_0 = 500 \cdot 500 = 250,000$, so

$$b(\mathbf{q}_0) = \frac{v}{n \ln n} \cdot n \cdot q_0 = \frac{v \cdot q_0}{\ln n} = \frac{2 \cdot 500}{\ln 500} \approx 160.9$$

3.4.3 Step 2: The Cost Function $C(\mathbf{q}_0)$

The ratio $q_0/b(\mathbf{q}_0)$ appears throughout the cost and price functions, so we compute it once:

$$\frac{q_0}{b(\mathbf{q}_0)} = \frac{q_0}{v \cdot q_0 / \ln n} = \frac{\ln n}{v} = \frac{\ln 500}{2} \approx 3.107$$

This ratio depends only on n and v , not on q_0 . It governs the “temperature” of the exponential terms inside the cost function: higher values compress the softmax toward uniformity, lower values allow sharper distributions.

The QLS-LMSR cost function is

$$C(\mathbf{q}) = 100 \cdot b(\mathbf{q}) \cdot \ln \left(\sum_{j=1}^n \exp \left(\frac{q_j}{b(\mathbf{q})} \right) \right)$$

At the uniform state every $q_j = q_0$, so the sum inside the logarithm collapses to n identical terms:

$$\sum_{j=1}^n \exp \left(\frac{q_0}{b(\mathbf{q}_0)} \right) = n \cdot \exp \left(\frac{q_0}{b(\mathbf{q}_0)} \right)$$

Therefore:

$$C(\mathbf{q}_0) = 100 \cdot b(\mathbf{q}_0) \cdot \ln \left(n \cdot \exp \left(\frac{q_0}{b(\mathbf{q}_0)} \right) \right) = 100 \cdot b(\mathbf{q}_0) \cdot \left[\ln n + \frac{q_0}{b(\mathbf{q}_0)} \right]$$

Substituting $b(\mathbf{q}_0) = v q_0 / \ln n$ and $q_0/b(\mathbf{q}_0) = \ln n / v$:

$$\begin{aligned} C(\mathbf{q}_0) &= 100 \cdot \frac{v q_0}{\ln n} \cdot \left[\ln n + \frac{\ln n}{v} \right] \\ &= 100 \cdot \frac{v q_0}{\ln n} \cdot \ln n \cdot \left[1 + \frac{1}{v} \right] \\ &= 100 \cdot q_0 \cdot (v + 1) \end{aligned}$$

The $\ln n$ terms cancel completely, yielding a closed form that depends only on the seed quantity and the sensitivity constant:

$$\boxed{C(\mathbf{q}_0) = 100 \cdot q_0 \cdot (v + 1) = 100 \cdot 500 \cdot 3 = 150,000 \text{ sats} = 0.00150000 \text{ BTC}}$$

This is the operator’s initial capital commitment—the total value locked in the cost function at the moment the market opens. It does not depend on n : whether the market has 100 bins or 1,000 bins, the startup cost is the same for a given q_0 and v .

3.4.4 Step 4: The Price Function $p_i(\mathbf{q}_0)$

The QLS-LMSR price function has two terms:

$$p_i(\mathbf{q}) = \underbrace{100 \cdot \alpha \cdot \ln \left(\sum_j \exp \left(\frac{q_j}{b(\mathbf{q})} \right) \right)}_{\text{Term 1}} + \underbrace{100 \cdot \frac{\sum_j q_j \cdot \exp(q_i/b) - \sum_j q_j \cdot \exp(q_j/b)}{(\sum_j q_j) \cdot (\sum_j \exp(q_j/b))}}_{\text{Term 2}}$$

Term 2 vanishes at the uniform state. When all $q_j = q_0$, the exponential $\exp(q_i/b)$ takes the same value for every index. The numerator of Term 2 becomes

$$\sum_j q_0 \cdot \exp(q_0/b) - \sum_j q_0 \cdot \exp(q_0/b) = 0$$

Term 2 captures the price correction arising from the fact that buying contract i increases the total quantity $\sum_j q_j$, which in turn increases the liquidity parameter b . At the uniform state, every contract has an identical relationship to b , so the correction is symmetric across all outcomes and vanishes.

Term 1 at the uniform state simplifies by the same algebraic cancellations as the cost function:

$$\begin{aligned} p_i(\mathbf{q}_0) &= 100 \cdot \alpha \cdot \ln \left(n \cdot \exp \left(\frac{q_0}{b(\mathbf{q}_0)} \right) \right) \\ &= 100 \cdot \frac{v}{n \ln n} \cdot \left[\ln n + \frac{\ln n}{v} \right] \\ &= 100 \cdot \frac{v}{n \ln n} \cdot \ln n \cdot \left(1 + \frac{1}{v} \right) \\ &= \frac{100(v+1)}{n} \end{aligned}$$

Again, $\ln n$ cancels entirely:

$$\boxed{p_i(\mathbf{q}_0) = \frac{100(v+1)}{n} = \frac{100 \cdot 3}{500} = 0.6 \text{ sats per contract}}$$

Every one of the 500 contracts has the same price at initialization. The full price vector is

$$\mathbf{p}(\mathbf{q}_0) = \underbrace{(0.6, 0.6, \dots, 0.6)}_{500} \in \mathbb{R}^{500}$$

3.4.5 Step 5: The Price Sum and Its Interpretation

In the standard (constant- b) LMSR, prices are the softmax and sum to exactly 100 sats (= 1 in the unit-payoff convention). In the liquidity-sensitive variant, Othman et al. [15] prove that

any market maker satisfying both path independence and liquidity sensitivity must relax the no-arbitrage condition $\sum_i p_i = 100$; prices necessarily sum to *at least* 100 and can sum to strictly more. The sum of all 500 contract prices at initialization is

$$\sum_{i=1}^n p_i(\mathbf{q}_0) = n \cdot \frac{100(v+1)}{n} = 100(v+1) = 300 \text{ sats}$$

The upper bound on the price sum is $100(1+v)$, and this maximum is achieved precisely when all quantities are equal—that is, at the uniform state. Why is the uniform state the worst case?

The price sum exceeds 100 sats because the uniform state spreads probability mass equally across all 500 bins, including outcomes that carry negligible true probability for any near-term settlement date. Othman et al. prove that any market maker satisfying both path independence and liquidity sensitivity must allow $\sum_i p_i \geq 100$ [15]. The excess reflects the cost of maintaining liquidity across the full outcome space. At the uniform state this cost is maximal. Restricting attention to the ~ 170 bins where realistic probability mass concentrates, the price sum over those bins alone would be approximately 100 sats.

The surplus is self-correcting. As traders concentrate purchases on likely bins, the quantity vector departs from uniformity and the price sum decreases toward 100 from above. The gap functions as the market maker’s effective spread: largest when uncertainty is highest, shrinking as the forecast sharpens. We do not restrict the outcome space to accelerate this convergence, because Bitcoin’s historical volatility—order-of-magnitude moves over multi-year horizons, and extreme tail events over shorter ones—demands comprehensive coverage from [\$0, \$500,000+).

3.4.6 Summary of Initial State

Quantity	Algebraic Form	Numerical Value
α	$v/(n \ln n)$	$\approx 6.44 \times 10^{-4}$
$b(\mathbf{q}_0)$	$v \cdot q_0 / \ln n$	≈ 160.9
$q_0/b(\mathbf{q}_0)$	$\ln n / v$	≈ 3.107
$C(\mathbf{q}_0)$	$100 \cdot q_0 \cdot (v+1)$	150,000 sats (0.00150 BTC)
$p_i(\mathbf{q}_0)$ for all i	$100(v+1)/n$	0.6 sats per contract
$\sum_i p_i(\mathbf{q}_0)$	$100(v+1)$	300 sats

Table 1: QLS-LMSR initial state under the default parameterization

4 Trading Strategies for Quantile Prediction Markets

The QLS-LMSR creates a well-defined arena for quantitative trading since prices are deterministic functions of the publicly observable state vector $\mathbf{q}$. All that remains is for traders is to bring an information advantage—a probability estimate more accurate than the market’s—and to deploy it profitably.

4.1 Truthful Trading is the Optimal Strategy

The most fundamental question a trader can ask about this market is: should I reveal my true beliefs, or can I do better by bluffing?

Chen et al. [8] answered this question definitively for the LMSR under the information structure relevant to financial forecasting. When traders hold *conditionally independent* signals—meaning each trader’s private information is drawn independently conditional on the true state of the world—the following result holds.

Theorem 1 (Unique Truthful Equilibrium; [8]). *In an LMSR with conditionally independent signals, truthful betting is the unique Weak Perfect Bayesian Equilibrium. No trader can improve their expected profit by bluffing, and traders prefer to act first.*

Conditional independence is the natural model for Bitcoin price forecasting. The future closing price P_T is a fixed but unknown quantity, and different traders estimate it using independent methods: one trader might use on-chain analytics, another might use their custom-built time series model based on the Bitcoin power law, another might use implied volatility from options prices. Their estimation errors are unrelated because they arise from different sources. Under this structure, the theorem says something powerful: the single best thing a trader can do is move the market price toward their honest belief as quickly as possible.

In an LMSR, the total profit available to all traders is fixed—it equals the subsidy provided by the market maker. A trader who reveals their information immediately captures the profit associated with that information. A trader who bluffs leaves profit on the table for others to take.

In practical terms, “truthful trading” on the QLS-LMSR means exactly this: if the market-implied probability for bin ω_i is $\hat{\pi}_i$ and your model says the true probability is higher, you buy contracts on ω_i , pushing the price up toward your estimate. If your model says the true probability is lower, you sell contracts on ω_i , pushing the price down. You profit when the market was wrong and your model was right.

4.2 Optimal Trade Sizing: The Kelly Criterion

Truthful trading tells the trader *what* to trade: buy bins the market underprices, sell bins it overprices. What it does not address is *how much* to trade—how many contracts to buy,

given a finite bankroll and the risk of loss. This is the problem that the Kelly criterion solves.

4.2.1 Terminal Wealth

Let B denote the trader's bankroll in sats. The trader's model produces a probability distribution $\boldsymbol{\pi}^* = (\pi_1^*, \dots, \pi_n^*)$ over the n quantile bins. The current market state $\mathbf{q}$ determines prices $\mathbf{p} = (p_1, \dots, p_n)$. The trader wishes to purchase a bundle $\boldsymbol{\Delta} = (\Delta_1, \dots, \Delta_n) \in \mathbb{Z}_{\geq 0}^n$, where Δ_i is the number of contracts on bin ω_i .

Because the cost function C is convex, the total cost of bundle $\boldsymbol{\Delta}$ starting from state $\mathbf{q}$ is

$$\text{Cost}(\boldsymbol{\Delta}) = C(\mathbf{q} + \boldsymbol{\Delta}) - C(\mathbf{q}),$$

which is *not* proportional to the number of contracts: each additional contract on a given bin pushes the price upward, making the next one more expensive. Including Glimpse's transaction fee at rate τ (default $\tau = 0.02$) on both purchases and redemptions, the trader's terminal wealth when outcome ω_j is realized is

$$W_j(\boldsymbol{\Delta}) = B - (1 + \tau)[C(\mathbf{q} + \boldsymbol{\Delta}) - C(\mathbf{q})] + (1 - \tau) \cdot 100 \cdot \Delta_j. \quad (1)$$

The three terms are: the starting bankroll, the total purchase cost (inflated by the fee), and the net payout of 100 sats per contract on the realized bin (reduced by the redemption fee). Contracts on every other bin expire worthless.

4.2.2 Stage 1: Identify Mispriced Bins

The market-implied probability of bin i is $\hat{\pi}_i = p_i(\mathbf{q})/100$. A bin is *underpriced* when the model assigns it a higher probability than the market does. The set of tradeable bins is

$$\mathcal{A} = \{i : \pi_i^* > \hat{\pi}_i\}.$$

For the remainder of this section we consider only purchases on underpriced bins; overpriced bins are handled symmetrically.

4.2.3 Stage 2: Compute the Truthful Quantity

For each bin $i \in \mathcal{A}$, the *truthful cap* $\Delta_i^{\max}$ is the number of contracts that would push the market price up to the model's probability:

$$p_i(\mathbf{q} + \Delta_i^{\max} \cdot \mathbf{e}_i) = 100 \cdot \pi_i^* \quad (2)$$

where $\mathbf{e}_i$ is the i -th standard basis vector. Beyond $\Delta_i^{\max}$, each additional contract costs more than it is worth under the trader's own model, so the expected profit per contract turns negative. The truthful cap can be found numerically by evaluating the price function at increasing values of Δ or by root-finding.

4.2.4 Stage 3: Kelly-Optimal Bet Size

The truthful cap bounds how many contracts are *worth* buying. A trader with a small bankroll may not be able to afford all of them. The Kelly criterion determines how many to buy by maximizing the expected logarithm of terminal wealth under the trader’s model:

$$\max_{\Delta \geq 0} \sum_{j=1}^n \pi_j^* \cdot \log W_j(\Delta) \quad (3)$$

subject to the budget constraint

$$(1 + \tau)[C(\mathbf{q} + \Delta) - C(\mathbf{q})] \leq B. \quad (4)$$

The reason for maximizing the *logarithm* of wealth rather than wealth itself is that wealth compounds multiplicatively. A trader who maximizes expected wealth directly will bet the entire bankroll every day, which guarantees eventual ruin: a single losing day erases everything. The logarithm goes to $-\infty$ as wealth approaches zero, which forces the optimizer to hold back enough of the bankroll to survive losing days. Over many repeated markets, maximizing expected log-wealth produces the fastest possible compound growth rate—this is the fundamental theorem behind the Kelly criterion [1].

4.2.5 Stage 4: Safety Margins and Execution

The Kelly-optimal quantities assume the model’s probabilities are exactly correct. In practice they are not. Benter (1994) [1] warns that if the trader overestimates the edge by a factor of two, full Kelly betting causes the bankroll to *shrink* over time rather than grow. Since estimation error almost always inflates the perceived edge, the risk of overbetting is severe and asymmetric.

The standard protection is *fractional Kelly*: scale every position by a constant $c \in (0, 1)$. Common choices are $c = 1/2$ (half Kelly) or $c = 1/3$ (third Kelly). Fractional Kelly has an elegant interpretation: a c -Kelly trader behaves identically to a full Kelly trader whose beliefs are a weighted average of the model and the market,

$$\tilde{\pi}_i = c \cdot \pi_i^* + (1 - c) \cdot \hat{\pi}_i.$$

A half-Kelly trader implicitly concedes that the market is half right. For a starting bankroll of $B = 0.05$ BTC, a conservative $c = 1/4$ is advisable during initial live trading.

The final execution quantity for each bin is capped by the truthful quantity and rounded down to a whole number of contracts:

$$\Delta_i^{\text{exec}} = \left\lfloor \min(c \cdot \Delta_i^{\text{Kelly}}, \Delta_i^{\text{max}}) \right\rfloor. \quad (5)$$

Rounding down ensures the trader never exceeds the Kelly recommendation. Bins where the floor is zero—the edge is too small to justify even one contract—are not traded.

4.2.6 Summary

Given a model distribution $\boldsymbol{\pi}^*$, market state $\mathbf{q}$, bankroll B , fee rate τ , and Kelly fraction c :

Step 1. Compute $\hat{\pi}_i = p_i(\mathbf{q})/100$ for each bin. Identify the underpriced set $\mathcal{A} = \{i : \pi_i^* > \hat{\pi}_i\}$.

Step 2. For each $i \in \mathcal{A}$, find $\Delta_i^{\max}$ by solving $p_i(\mathbf{q} + \Delta_i^{\max} \cdot \mathbf{e}_i) = 100 \cdot \pi_i^*$.

Step 3. Solve the Kelly optimization (3) subject to (4) to obtain Δ_i^{Kelly} for each $i \in \mathcal{A}$.

Step 4. Set $\Delta_i^{\text{exec}} = \lfloor \min(c \cdot \Delta_i^{\text{Kelly}}, \Delta_i^{\max}) \rfloor$ and execute the trade.

5 Worked Example: Three Traders Build a Forecast

We continue from the initial state computed in Section 3.4. Suppose Bitcoin is trading around \$80,000, and three traders arrive in sequence with different views: Alice hedges a bearish scenario, Bob expresses a bullish bet, and Quigley deploys a quantitative model. Walking through their trades illustrates how the QLS-LMSR aggregates conflicting beliefs into a single forecast—and how even modest trades from ordinary participants can give the market meaningful structure.

5.1 The Starting State

Before any trades, every bin is priced at 0.6 sats and the market conveys no information. Figure 1 shows the price histogram over the range [\$59,000, \$99,000): a flat line at 0.6 sats across every bin. This is the uniform prior, and it is identical at every other point on the price axis from [\$0, \$1,000) through [\$499,000, ∞).

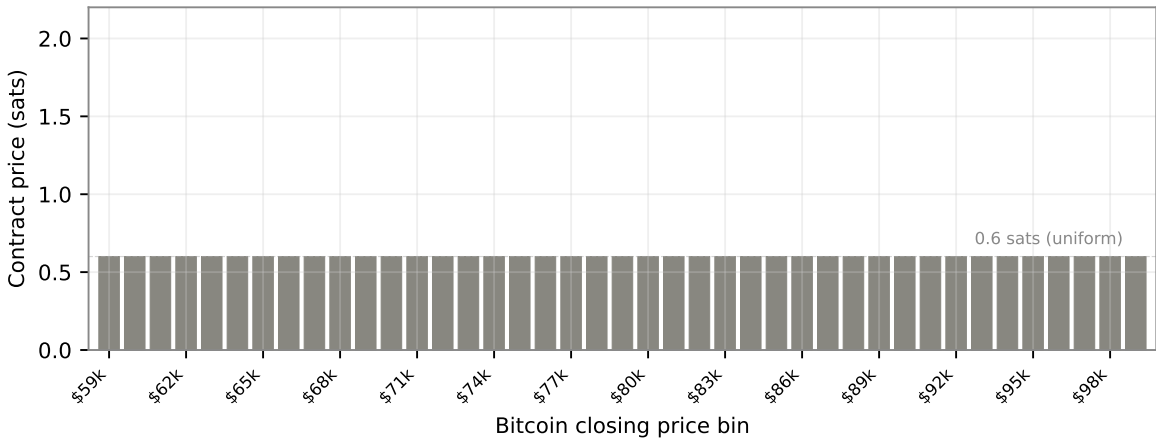


Figure 1: Initial market state: all 500 bins priced uniformly at 0.6 sats. The market has no informational content.

5.2 Alice: A Bearish Hedge

Alice holds Bitcoin and is worried about a near-term decline. She wants to hedge her exposure by purchasing contracts covering the range [\$74,000, \$81,000)—seven bins that span a scenario where Bitcoin stays flat or drops. Specifically, she buys $\Delta = 200$ contracts on each of bins ω_{75} through ω_{81} , for a total of 1,400 contracts.

Cost. At the uniform initial state $\mathbf{q}_0$, Alice’s trade moves the quantity on each of her seven bins from 500 to 700. The cost is the difference in the cost function:

$$\text{Cost}_{\text{Alice}} = C(\mathbf{q}_1) - C(\mathbf{q}_0) = 151,104 - 150,000 = 1,104 \text{ sats}$$

Including the 2% transaction fee, Alice pays $(1.02)(1,104) \approx 1,126$ sats, or about 0.00001126 BTC. At \$80,000 per BTC, this is roughly \$0.90—less than a cup of coffee.

Price impact. Figure 2 shows the market after Alice’s trade. The liquidity parameter increases from $b \approx 160.9$ to $b \approx 161.8$, reflecting the additional contracts in the system. Prices on Alice’s seven bins rise from 0.60 sats to 1.06 sats, while every other bin dips slightly to 0.59 sats. The redistribution is intuitive: Alice has injected information (“I think this range is more likely than the market does”), and the cost function responds by raising prices on those bins and lowering prices elsewhere.

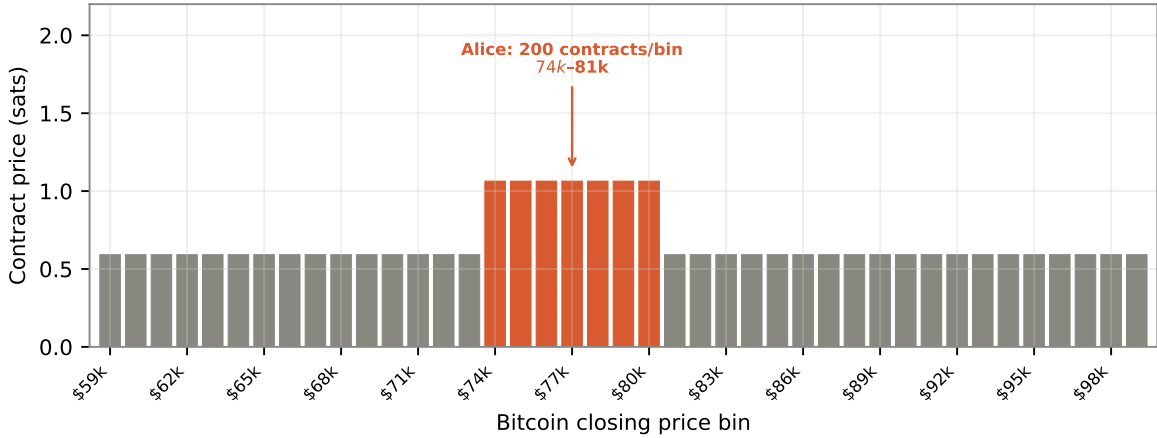


Figure 2: Market state after Alice buys 200 contracts on each of bins ω_{75} – ω_{81} (orange). Her bins nearly double in price to 1.06 sats; all other bins dip to 0.59 sats. Total cost: 1,126 sats ($\approx$ \$0.90).

Payoff structure. Each of Alice’s bins contains 200 contracts. If Bitcoin closes in any one of those bins, Alice receives $(1 - \tau) \times 100 \times 200 = 19,600$ sats, for a net profit of $19,600 - 1,126 = 18,474$ sats—a return of roughly $17.4\times$ her cost, or $+1,641\%$. If Bitcoin closes outside her range, she loses the 1,126 sats she paid. Because she covers seven bins, Alice needs only one of them to hit, so the effective break-even probability across the full range is $1,126/19,600 \approx 5.7\%$ —meaning Alice profits as long as she believes there is at least a 5.7% chance Bitcoin closes between \$74,000 and \$81,000.

5.3 Bob: A Bullish Bet

Bob arrives next. He is bullish: he believes Bitcoin will either hold steady or rally, and he is willing to accept a scenario where it dips slightly but not as far as Alice fears. He buys $\Delta = 500$ contracts on each of bins ω_{80} through ω_{86} , covering $[\$79,000, \$86,000]$ —a total of 3,500 contracts.

Cost. Bob trades into the market *after* Alice, so he faces the post-Alice state $\mathbf{q}_1$:

$$\text{Cost}_{\text{Bob}} = C(\mathbf{q}_2) - C(\mathbf{q}_1) = 158,673 - 151,104 = 7,569 \text{ sats}$$

With the 2% fee, Bob pays $(1.02)(7,569) \approx 7,721$ sats (≈ 0.00007721 BTC $\approx \$6.18$). Bob’s trade costs substantially more than Alice’s, for two reasons: he buys more contracts per bin (500 vs. 200), and two of his bins (ω_{80} and ω_{81}) already had elevated prices from Alice’s prior trade. The convexity of the cost function means that buying into bins where quantity has already accumulated is progressively more expensive—this is the mechanism by which the market charges more for reinforcing existing beliefs and less for contributing novel information.

Bin Range	Quantity	Price (sats)	Trader(s)
Background (all other bins)	500	0.48	—
[\$74,000, \$79,000)	700	0.79	Alice only
[\$79,000, \$81,000)	1,200	9.81	Alice + Bob
[\$81,000, \$86,000)	1,000	3.14	Bob only

Table 2: Contract prices after both trades. The overlap zone commands the highest price because both traders independently agree it is likely.

Price impact. Figure 3 shows the market after both trades. The distribution now has clear structure:

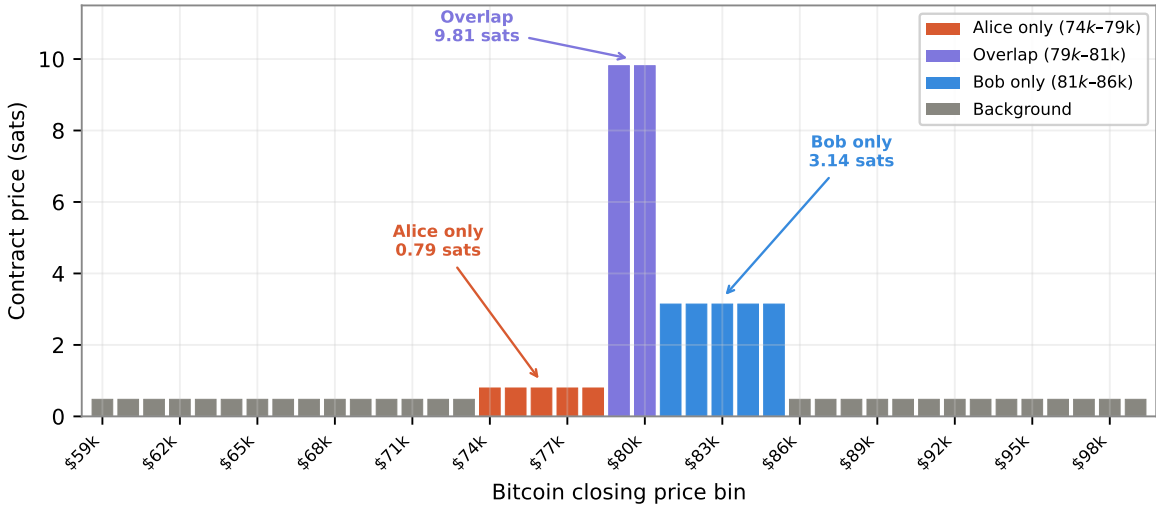


Figure 3: Market state after both Alice and Bob have traded. The overlap zone $[\$79,000, \$81,000)$ (purple) dominates at 9.81 sats—both traders independently agree this range is likely. Alice’s exclusive range $[\$74,000, \$79,000)$ (orange) sits at 0.79 sats, Bob’s exclusive range $[\$81,000, \$86,000)$ (blue) at 3.14 sats, and all other bins (gray) have fallen to 0.48 sats.

The overlap zone $[\$79,000, \$81,000)$ stands out: at 9.81 sats per contract, these bins have risen more than $16\times$ from the initial 0.60. This is the market aggregating two independent signals into a region of consensus. The total price sum has fallen from 300 sats at initialization to 272.6 sats, reflecting the market’s increased confidence—probability mass is concentrating, and the effective spread is shrinking.

Each of Bob’s bins contains 500 contracts. If Bitcoin closes in any one of them, he receives $(1 - \tau) \times 100 \times 500 = 49,000$ sats, for a net profit of $49,000 - 7,721 = 41,279$ sats (+535%). His break-even probability across the full seven-bin range is $7,721/49,000 \approx 15.8\%$ —he needs to believe there is at least a 15.8% chance Bitcoin closes anywhere in $[\$79,000, \$86,000)$ to justify the trade.

5.3.1 Settlement Scenarios for Alice and Bob

Three settlement outcomes illustrate the range of results:

Scenario 1: Bitcoin closes at \$80,500 (bin ω_{81} , in the overlap zone). Both traders win. Alice nets +18,474 sats; Bob nets +41,279 sats. The market maker pays out both positions from the cost function.

Scenario 2: Bitcoin rallies to \$85,500 (bin ω_{86} , Bob only). Alice loses her 1,126 sats—but she still holds her Bitcoin, which has appreciated. Bob nets +41,279 sats.

Scenario 3: Bitcoin crashes to \$75,500 (bin ω_{76} , Alice only). Alice nets +18,474 sats—her hedge pays off, partially offsetting the decline in her Bitcoin holdings. Bob loses his 7,721 sats.

In every scenario, each trader’s maximum loss is exactly what they paid upfront. There is no margin, no liquidation, and no counterparty risk. The cost function guarantees settlement with a bounded loss from the market maker.

5.4 Quigley the Quant Enters the Market

Alice and Bob traded on intuition—bearish fear and bullish conviction. Quigley trades on a model. He runs a proprietary ensemble model that combines on-chain metrics, market microstructure data, and macroeconomic features to produce a conditional density forecast. The pipeline outputs a full probability distribution π^* over all 500 quantile bins.

Quigley’s bankroll is $B = 0.05 \text{ BTC} = 5,000,000$ sats—roughly \$4,000 at current prices. His plan is to trade daily, compounding returns over time, with quarter-Kelly sizing ($c = 1/4$) to guard against model overconfidence.

5.4.1 Quigley’s Model Distribution

Suppose Quigley’s ensemble model, conditioned on today’s feature set, produces the distribution shown in Table 3. His model is more bullish than the current market consensus: it places the mode at ω_{83} ([\\$82,000, \\$83,000)) with 21% probability, and concentrates 99% of its mass across 13 bins from ω_{77} through ω_{89} . The remaining 1% is spread across all 487 other bins as a diffuse tail.

Bin	Price Range	Model π_i^*	Market $\hat{\pi}_i$
ω_{77}	[\$76,000, \$77,000)	1.00%	0.79%
ω_{78}	[\$77,000, \$78,000)	2.00%	0.79%
ω_{79}	[\$78,000, \$79,000)	4.00%	0.79%
ω_{80}	[\$79,000, \$80,000)	7.00%	9.81%
ω_{81}	[\$80,000, \$81,000)	11.00%	9.81%
ω_{82}	[\$81,000, \$82,000)	17.00%	3.14%
ω_{83}	[\$82,000, \$83,000)	21.00%	3.14%
ω_{84}	[\$83,000, \$84,000)	16.00%	3.14%
ω_{85}	[\$84,000, \$85,000)	10.00%	3.14%
ω_{86}	[\$85,000, \$86,000)	5.00%	3.14%
ω_{87}	[\$86,000, \$87,000)	3.00%	0.48%
ω_{88}	[\$87,000, \$88,000)	1.50%	0.48%
ω_{89}	[\$88,000, \$89,000)	0.50%	0.48%
All others	—	$\sim 0.002\%$ each	0.48%

Table 3: Quigley’s model distribution π^* compared with the market-implied distribution $\hat{\pi}$ after Alice’s and Bob’s trades. The model is substantially more bullish than the market, placing 64% of its mass above \$81,000.

Two features of this comparison are immediate. First, Quigley’s model agrees with Alice and Bob that extreme outcomes are unlikely—nearly all mass falls within a \$13,000 window—but *disagrees* about where within that window the price will land. Second, the model identifies ω_{80} ([\\$79,000, \$80,000)) as *overpriced*: Alice’s and Bob’s overlapping purchases have driven it to 9.81%, but Quigley’s model assigns only 7%. The rational response is not to trade that bin at all.

5.4.2 Step 1: Identify Mispriced Bins

Quigley computes the edge $\pi_i^* - \hat{\pi}_i$ for every bin. The underpriced set is

$$\mathcal{A} = \{i : \pi_i^* > \hat{\pi}_i\} = \{\omega_{77}, \omega_{78}, \omega_{79}, \omega_{81}, \omega_{82}, \omega_{83}, \omega_{84}, \omega_{85}, \omega_{86}, \omega_{87}, \omega_{88}\}$$

Eleven bins are underpriced. The edges range from +0.21% (ω_{77} , a marginal opportunity) to

+17.86% (ω_{83} , where Quigley’s model assigns 21% to a bin the market prices at 3.14%). Bin ω_{80} is excluded: the market overprices it relative to the model.

Figure 4 makes the mismatch visual. The market-implied distribution (purple) is dominated by the Alice-Bob overlap at [\$79k, \$81k), while Quigley’s model (green) concentrates mass several thousand dollars higher. The gap between the green and purple bars on bins ω_{82} – ω_{84} is the trading opportunity; the purple bar towering over the green on ω_{80} is the overpriced bin he avoids.

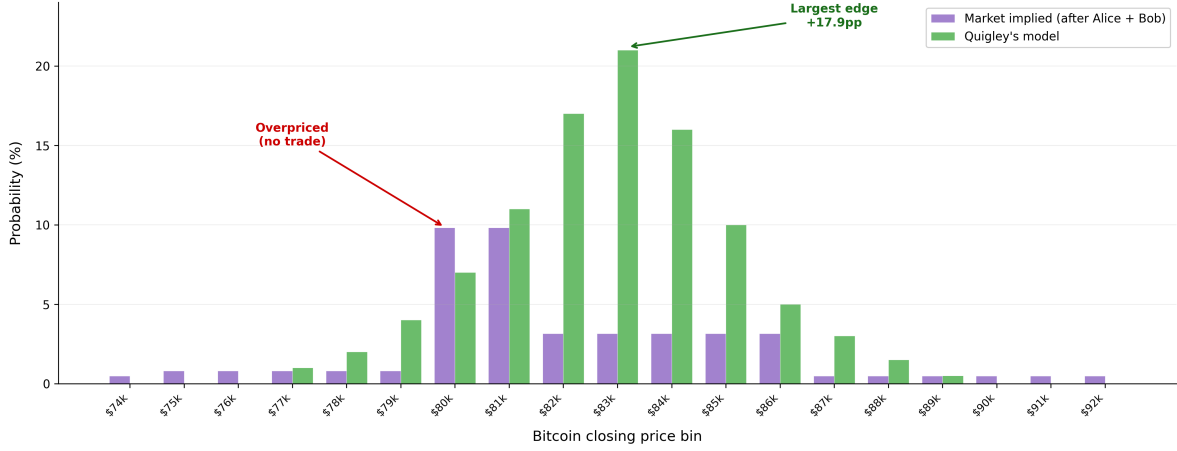


Figure 4: Quigley’s model distribution (green) versus the market-implied distribution after Alice and Bob (purple), zoomed to the active range [\$74k, \$92k). The model’s mode at ω_{83} (21%) dwarfs the market’s 3.14% implied probability—an edge of +17.9 percentage points. Bin ω_{80} (marked “Overpriced”) is the only bin where the market assigns *higher* probability than the model; Quigley does not trade it.

5.4.3 Step 2: Compute Truthful Caps

For each underpriced bin, Quigley solves equation (2) numerically to find the truthful cap $\Delta_i^{\max}$ —the quantity that would push the market price up to his model’s probability. Table 4 reports the results.

Bin	Range	Edge	$\Delta_i^{\max}$	Target Price	Cost Alone
ω_{77}	[\$76,000, \$77,000)	+0.21%	62	1.00 sats	55 sats
ω_{78}	[\$77,000, \$78,000)	+1.21%	216	2.00 sats	275 sats
ω_{79}	[\$78,000, \$79,000)	+3.21%	349	4.00 sats	656 sats
ω_{81}	[\$80,000, \$81,000)	+1.19%	21	11.00 sats	218 sats
ω_{82}	[\$81,000, \$82,000)	+13.86%	319	17.00 sats	2,643 sats
ω_{83}	[\$82,000, \$83,000)	+17.86%	362	21.00 sats	3,457 sats
ω_{84}	[\$83,000, \$84,000)	+12.86%	306	16.00 sats	2,429 sats
ω_{85}	[\$84,000, \$85,000)	+6.86%	215	10.00 sats	1,270 sats
ω_{86}	[\$85,000, \$86,000)	+1.86%	86	5.00 sats	343 sats
ω_{87}	[\$86,000, \$87,000)	+2.52%	496	3.00 sats	591 sats
ω_{88}	[\$87,000, \$88,000)	+1.02%	356	1.50 sats	291 sats

Table 4: Truthful caps for Quigley’s underpriced bins. The “Cost Alone” column shows what it would cost to buy $\Delta_i^{\max}$ contracts on that single bin, holding all other bins constant. The actual multi-bin trade will cost less per bin due to the shared liquidity effect.

On ω_{83} —the bin with the largest edge—Quigley could purchase up to 362 contracts before the market price reaches his target of 21.00 sats. But these caps represent the *maximum* quantity worth buying under his model. A trader with a finite bankroll and healthy respect for model uncertainty will buy far fewer.

5.4.4 Step 3: Kelly-Optimal Allocation

Quigley solves the multi-outcome Kelly optimization (3) subject to the budget constraint (4), using his model distribution π^* and bankroll $B = 5,000,000$ sats. The optimizer allocates contracts proportionally to each bin’s contribution to expected log-wealth, which depends on the product of the edge (how mispriced the bin is) and the model probability (how likely the bin is to pay off).

At full Kelly ($c = 1$), the optimizer recommends a total expenditure of approximately 10,221 sats—just 0.2% of Quigley’s bankroll. This may seem surprisingly conservative, but it reflects two features of the QLS-LMSR environment: the market is still very young (only two prior traders), so the liquidity parameter b is low and prices move steeply with each contract purchased; and the convexity of the cost function means that buying many contracts on a single bin rapidly becomes expensive. The Kelly criterion automatically accounts for this nonlinearity.

5.4.5 Step 4: Fractional Kelly Execution

Quigley applies quarter-Kelly scaling ($c = 1/4$) because his model is new and untested in live markets. Following equation (5), each full-Kelly quantity is multiplied by 0.25, capped at the

truthful quantity, and rounded down. The executed trade is shown in Table 5.

Bin	Range	π_i^*	Edge	Δ_i^{Kelly}	$c \cdot \Delta_i^{\text{Kelly}}$	Δ_i^{exec}
ω_{77}	[\$76k, \$77k)	1.0%	+0.21%	0.3	0.1	0
ω_{78}	[\$77k, \$78k)	2.0%	+1.21%	3.1	0.8	0
ω_{79}	[\$78k, \$79k)	4.0%	+3.21%	16.6	4.1	4
ω_{81}	[\$80k, \$81k)	11.0%	+1.19%	16.9	4.2	4
ω_{82}	[\$81k, \$82k)	17.0%	+13.86%	304.7	76.2	76
ω_{83}	[\$82k, \$83k)	21.0%	+17.86%	485.0	121.2	121
ω_{84}	[\$83k, \$84k)	16.0%	+12.86%	266.0	66.5	66
ω_{85}	[\$84k, \$85k)	10.0%	+6.86%	88.7	22.2	22
ω_{86}	[\$85k, \$86k)	5.0%	+1.86%	12.0	3.0	3
ω_{87}	[\$86k, \$87k)	3.0%	+2.52%	9.8	2.4	2
ω_{88}	[\$87k, \$88k)	1.5%	+1.02%	2.0	0.5	0
Total:						298

Table 5: Quigley’s quarter-Kelly execution. Bins ω_{77} , ω_{78} , and ω_{88} round to zero contracts and are not traded: the edge exists but is too small to justify even one contract at this Kelly fraction.

The allocation is bell-shaped: ω_{83} receives the most contracts (121) because it has both the highest model probability (21%) and the largest edge (+17.86%), while tail bins receive few or zero. Three underpriced bins (ω_{77} , ω_{78} , ω_{88}) are dropped entirely after quarter-Kelly scaling rounds them to zero—a tiny edge on a low-probability bin does not warrant even one contract at this confidence level.

The total bet is strikingly small. The 298 contracts across 8 bins cost 1,178 sats before fees, or 1,201 sats after the 2% transaction fee—0.024% of Quigley’s bankroll, roughly \$0.96. The Kelly criterion is often described as “aggressive,” but at quarter-Kelly in a young, low-liquidity market, it is astonishingly conservative. A trader who bets 0.024% per day can sustain over 4,000 consecutive losing days before depleting the bankroll—ample runway to survive model miscalibration, regime changes, and bad luck.

5.4.6 The Trade’s Cost and the Market’s Response

Quigley’s trade costs a total of **1,201 sats** (including the 2% fee), leaving him with 4,998,799 sats—99.976% of his starting bankroll. After the trade, the market’s price distribution shifts noticeably:

Bin	Range	Before (sats)	After (sats)	Change
ω_{79}	[\$78k, \$79k)	0.79	0.77	−3%
ω_{80}	[\$79k, \$80k)	9.81	9.17	−7%
ω_{81}	[\$80k, \$81k)	9.81	9.39	−4%
ω_{82}	[\$81k, \$82k)	3.14	4.49	+43%
ω_{83}	[\$82k, \$83k)	3.14	5.80	+85%
ω_{84}	[\$83k, \$84k)	3.14	4.25	+35%
ω_{85}	[\$84k, \$85k)	3.14	3.33	+6%
Background	all other bins	0.48	0.46	−4%

Table 6: Price impact of Quigley’s trade. His purchases shift the market’s center of gravity upward from the [\$79k, \$81k) overlap zone toward [\$82k, \$83k), where his model places the mode.

The price sum falls from 272.6 sats to 268.4 sats—the market is becoming more confident as a third independent trader contributes information. Bin ω_{83} has nearly doubled in price from 3.14 to 5.80 sats, emerging as the new modal bin. The overlap zone [\$79k, \$81k), which Alice and Bob inflated to 9.81 sats, has declined to 9.17–9.39 sats: Quigley’s model disagrees that this range deserves such high weight, and by concentrating his purchases elsewhere, he has diluted the Alice-Bob consensus.

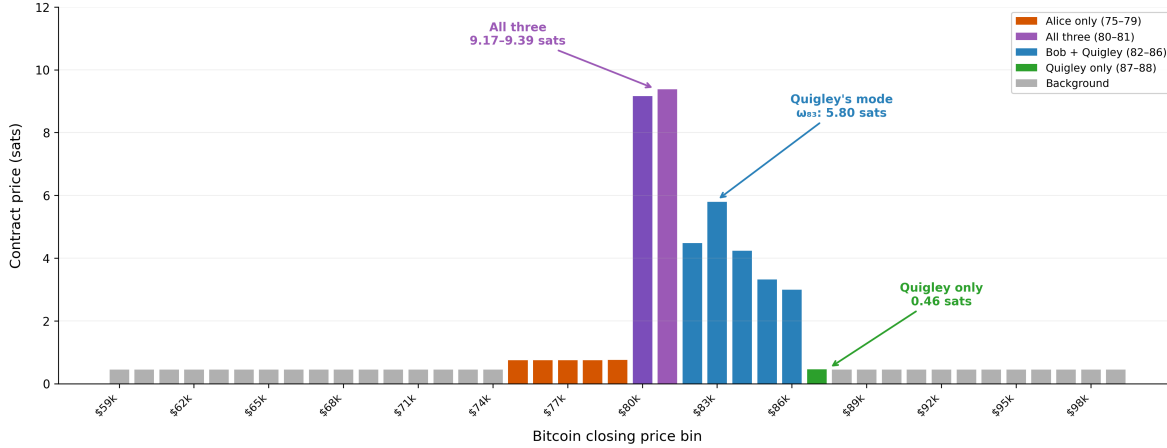


Figure 5: Market state after all three traders. The purple bins [\$79k, \$81k) reflect positions held by Alice, Bob, and Quigley; the blue bins [\$81k, \$86k) combine Bob’s and Quigley’s purchases into a bell-shaped shoulder peaking at ω_{83} (5.80 sats); the green bins [\$86k, \$88k) are Quigley’s tail positions alone. Compared to Figure 3, the market’s center of gravity has shifted upward from the overlap zone toward Quigley’s model mode, and the background has compressed from 0.48 to 0.46 sats as the forecast sharpens.

5.4.7 Quigley’s Payoff Scenarios

Table 7 shows Quigley’s profit or loss for each possible settlement bin.

Bin	Range	Δ_i^{exec}	Payout	Net P/L	Return
ω_{79}	[\$78k, \$79k)	4	392 sats	−809 sats	−67%
ω_{81}	[\$80k, \$81k)	4	392 sats	−809 sats	−67%
ω_{82}	[\$81k, \$82k)	76	7,448 sats	+6,247 sats	+520%
ω_{83}	[\$82k, \$83k)	121	11,858 sats	+10,657 sats	+887%
ω_{84}	[\$83k, \$84k)	66	6,468 sats	+5,267 sats	+438%
ω_{85}	[\$84k, \$85k)	22	2,156 sats	+955 sats	+80%
ω_{86}	[\$85k, \$86k)	3	294 sats	−907 sats	−76%
ω_{87}	[\$86k, \$87k)	2	196 sats	−1,005 sats	−84%
Any other bin	—	0	0 sats	−1,201 sats	−100%

Table 7: Quigley’s payoff by settlement outcome. Payout is $(1 - \tau) \times 100 \times \Delta_i^{\text{exec}}$; Net P/L is payout minus total cost of 1,201 sats.

The payoff structure is highly asymmetric. If Bitcoin closes in ω_{83} ([\$82,000, \$83,000)), Quigley’s 121 contracts pay 11,858 sats on an outlay of 1,201 sats—a return of +887%. If Bitcoin closes in *any* of ω_{82} – ω_{85} , he earns a net profit; under his model, these four bins carry a combined probability of 64%. If Bitcoin closes in ω_{79} , ω_{81} , ω_{86} , or ω_{87} —bins where he holds contracts but too few to recoup the total cost—he loses most of his outlay. And if Bitcoin closes in any untouched bin, he loses the full 1,201 sats. The maximum loss in every scenario is exactly what he paid upfront.

Under Quigley’s own model, the expected payout across all outcomes is 5,086 sats, yielding an expected profit of +3,885 sats (+324%) on a single daily market. If this expected return persists and his model is well-calibrated, the compound effect over many daily markets is substantial, even at quarter-Kelly.

5.4.8 The Discipline of the Kelly Criterion

It is instructive to compare what Quigley *actually* bets with what a naive trader might do. A trader who believes Bitcoin will close in [\$82,000, \$83,000) with 21% probability might be tempted to buy hundreds of contracts on that single bin, or to spread 5% of bankroll across a few favorites. The Kelly criterion says: no. The mathematically optimal bet is 1,201 sats—not because the edge is small, but because the market is thin (low b , so each additional contract is progressively more expensive).

If Quigley trades daily for a year and his model is accurate, the Kelly criterion guarantees that his compounding trajectory is, in expectation, the fastest possible among all strategies that do not risk ruin. This is the core promise of the framework: a disciplined pipeline from model

to trade, where every parameter—the edge, the quantity, the fraction, the cost—is determined by mathematics rather than intuition.

5.5 What Three Traders Have Built

Three participants have now traded. Alice spent 1,126 sats hedging a bearish view. Bob spent 7,721 sats expressing a bullish one. Quigley spent 1,201 sats implementing a model-driven forecast. The combined expenditure is 10,048 sats—about \$8 at current prices—and the market’s probability distribution has evolved from a flat, uninformative uniform prior into a structured forecast with a clear mode near \$82,000, elevated shoulders reflecting disagreement about direction, and a compressed background reflecting consensus that extreme outcomes are unlikely.

Notice the mechanism at work. Alice, Bob, and Quigley disagree—bearish, bullish, model-driven—yet the market does not choose one view over the other. It computes a *weighted synthesis*: the region where multiple traders agree gets the highest weight, and regions where only one trader has conviction get intermediate weight. Each trader contributes information, and the cost function synthesizes these signals into a single probabilistic object that is more informative than any individual view. The forecast is the product, and every subsequent trade refines it further.

6 Conclusion

We have presented the mechanism design of a quantile prediction market for Bitcoin built on the liquidity-sensitive logarithmic market scoring rule. The QLS-LMSR partitions the price axis into n bins and runs a single automated market maker across all of them, producing a probability distribution over the full outcome space that updates with every trade. The operator’s worst-case loss is bounded and known in advance, liquidity scales with participation, and prices are path-independent—there is nothing to spoof and no order book to game.

We showed that under the conditionally independent signal structure natural to financial forecasting, truthful trading is the unique equilibrium: no trader benefits from bluffing, and the rational move is to reveal your information immediately. We then sketched out a trading pipeline so that a trader with a probability model and a bankroll has everything needed to size and execute positions.

The argument running through all of this is that quantile prediction markets are a better structure for financial forecasting than either binary markets or limit-order-book exchanges. Binary markets capture one bit of information per contract. Order books are zero-sum, require matched counterparties, and reward adversarial behavior. The QLS-LMSR is positive-sum by construction, operates over the entire outcome space in a single liquidity pool, and turns every trade into a contribution to a shared probabilistic forecast.

7 Proper scoring rules elicit truthful beliefs

Consider a binary event with outcomes $\Omega = \{Y, N\}$. A forecaster has a true belief

$$p = \Pr(Y) \in (0, 1), \quad \Pr(N) = 1 - p,$$

but reports a probability $r \in (0, 1)$ for Y (and $1 - r$ for N). A scoring rule specifies payoffs $s_Y(r)$ if Y occurs and $s_N(r)$ if N occurs. Under risk neutrality, the forecaster chooses r to maximize expected score

$$\mathbb{E}_p[s(r)] := p s_Y(r) + (1 - p) s_N(r).$$

We illustrate properness with the logarithmic score. Let

$$s_Y(r) = \log r, \quad s_N(r) = \log(1 - r).$$

Then

$$\mathbb{E}_p[s(r)] = p \log r + (1 - p) \log(1 - r),$$

and

$$\frac{d}{dr} \mathbb{E}_p[s(r)] = \frac{p}{r} - \frac{1 - p}{1 - r}.$$

Setting the derivative to zero gives $p(1 - r) = (1 - p)r$, hence $r^* = p$. The forecaster maximizes expected score by reporting honestly.

By contrast, a non-proper rule can reward coarse, overconfident reporting. Consider the rule

$$s_Y(r) = \mathbf{1}\{r \geq 1/2\}, \quad s_N(r) = \mathbf{1}\{r \leq 1/2\},$$

which pays 1 if the realized outcome was assigned probability at least $1/2$ and 0 otherwise. The expected score becomes

$$\mathbb{E}_p[s(r)] = \begin{cases} p, & r > 1/2, \\ 1 - p, & r < 1/2, \end{cases}$$

(with any tie convention at $r = 1/2$). If $p > 1/2$, then every report $r > 1/2$ yields the same expected score p , so the rule fails to elicit the correct probability magnitude and only elicits which side of $1/2$ the forecaster is on.

Under a proper scoring rule, the report r is uniquely pinned down by first-order optimality at the true belief p . Under a non-proper rule, expected payoff can be flat over wide regions of reports, so the mechanism does not discipline probability reports toward truthfulness.

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